

Semantic Clustering and Classification of Food Preparation Protocols using Contextual Embeddings

Members:

DHAIRYA (2022157)

HARSH VISHWAKARMA (2022205)

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Science and Engineering**

BTP Track: Research

**BTP Advisor(s)
Ganesh Bagler**

**Indraprastha Institute of Information Technology,
Delhi**

Student's Declaration

I hereby declare that the work presented in the report entitled "Semantic Clustering and Classification of Food Preparation Protocols using Contextual Embeddings", submitted by me for the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science & Engineering at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under the guidance of Ganesh Bagler. Due acknowledgements have been given in the report to all material used. This work has not been submitted anywhere else for the award of any other degree.

Student Name

Harsh Vishwakarma
Dhairya

Place : New Delhi

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Advisor Name
Prof. Ganesh Bagler

Acknowledgement

I want to extend my heartfelt thanks to my guide, Prof. Ganesh Bagler. His guidance, knowledge, and constant encouragement made this project possible. I am grateful to the IIIT-Delhi faculty, research scholars, and the Complex System Lab members for their datasets.

Introduction

1.1 Background and Motivation

Food processing and preparation methods have profound implications for nutrition and public health. In recent years, food classification systems have been developed to capture the degree of processing in foods and diets. For example, the NOVA framework categorizes foods into four groups – unprocessed/minimally processed, processed culinary ingredients, processed foods, and ultra-processed foods (UPFs) – based on processing extent and intent . UPFs (NOVA4) are generally defined as “industrial formulations typically with five or more and usually many ingredients,” often containing additives not used in home cooking . Diets high in UPFs have been linked to adverse health outcomes, including obesity, diabetes, and heart disease . For instance, studies show that greater UPF intake tends to increase sugar and decrease fiber consumption .

Despite this concern, assigning foods or recipes to processing-based categories can be ambiguous and inconsistent. The NOVA definitions involve mixed technological and socio-cultural criteria that lack clear decision rules . Consequently, different classification schemes (NOVA, UNC, IFIC, IARC, etc.) can produce widely varying results when applied to the same data . This motivates the need for more rigorous, data-driven methods to analyze and classify culinary preparations.

1.2 Ultra-Processed Foods (UPF) Context

Ultra-processed foods are characterized by complex formulations and the presence of additives not typically used in home cooking. Diets high in UPFs have been linked to poor nutritional quality, including increased sugar intake and reduced fiber consumption. However, existing classification frameworks often yield inconsistent results when

applied to the same dataset, highlighting the need for alternative perspectives.

1.3 Challenges in Food Classification

Traditional food classification systems rely heavily on ingredient composition and processing intent, which can be subjective. Multiple frameworks often disagree in categorizing the same food items, leading to inconsistencies in research findings and policy recommendations.

1.4 Research Scope

In this study, we investigate an alternative dimension of food processing: the culinary protocols (cooking actions) described in recipes. A food preparation protocol is defined as a single cooking action or operation (a verb) extracted from recipe instructions (e.g. chop, mix, boil). We ask whether embedding-based semantic clustering of these protocols can reveal meaningful groups (e.g. transformation vs. mixing actions) that reflect their role in food preparation and processing. This approach is complementary to ingredient- or formulation-based classification and could help characterize processing patterns in culinary practice.

PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Traditional food classification systems (e.g. NOVA) focus on ingredients and formulation to assess processing level, but they can be subjective and inconsistent. Meanwhile, recipes also encode how foods are prepared through sequences of cooking actions (protocols). Yet, there is no standardized classification of these culinary actions in terms of their semantic roles or processing impact. This gap hampers

our ability to analyze recipes and diets at a finer granularity. The specific problem is: Can we automatically group cooking-action verbs (food protocols) into meaningful semantic categories using NLP embeddings, and thereby derive an objective, data-driven taxonomy of food preparation actions?

2.2 Research Questions

- What are the latent semantic categories of cooking actions?
- Which embedding model best captures protocol semantics?
- How well do clustering results align with human-labeled categories?

2.3 Objectives of the Study

- Extract cooking-action verbs from recipes.
- Create a manually labeled ground truth dataset.
- Generate embeddings using multiple models.
- Perform clustering of protocols.
- Evaluate clustering performance using ARI and silhouette score.
- Analyze cluster structures through visualization.

RELATED WORK

3.1 Food Processing Classification Studies

Many food classification systems have been proposed to describe processing level. The most widely used is NOVA, introduced by Monteiro et al., which categorizes foods into four groups by processing purpose . NOVA Group 4 (ultra-processed) are “industrial formulations...with five or more and usually many ingredients” often containing additives absent from traditional cuisine . Multiple studies have linked high UPF consumption to poor diet quality and health risks (e.g. obesity, cardiovascular disease, diabetes) . For instance, Pereira

et al. report that many chronic diseases correlate with diets rich in highly processed foods . \

However, the literature also highlights limitations of existing systems. Others have noted that NOVA's criteria are multidimensional and sometimes contradictory . Reviews by Moubarac et al. and Sadler et al. stress that different classification frameworks (NOVA, IARC, IFIC, UNC, etc.) often disagree on labeling the same foods . For example, a comparative study of five processing-based systems found up to 35–47 percentage-point differences in estimated UPF contributions . These inconsistencies motivate more objective, transparent methods of classifying foods and recipes by processing.

3.2 NLP and Embedding Models in Food Domain

Recent advances in NLP have enabled semantic representation of text using embedding models. Models such as SBERT, BioBERT, and domain-specific variants like FoodBERT have been applied to various food-related tasks including recipe understanding and ingredient analysis.

3.3 Research Gaps

Existing research focuses primarily on ingredients and recipes rather than cooking actions. This work addresses the gap by analyzing semantic relationships between cooking protocols.

DATASET AND PREPROCESSING

4.1 Dataset Description

Our analysis uses RecipeDB , a large curated recipe database developed by the IIIT-Delhi Complex Systems Lab . RecipeDB contains 118,171 recipes from around the world (6 continents, 74

countries), encompassing diverse cuisines . Each recipe entry includes metadata (title, region, dietary tags), nutritional information (calories, macros, etc.), ingredients, and preparation details. Importantly, the dataset has a “Processes” field that lists textual cooking steps (verbs and phrases) used in the recipe (e.g. “chop onions,” “simmer sauce”). Across all recipes, there are 268 distinct process terms (e.g. heat, bake, roast, etc.)

4.2 Data Attributes and Features

Key fields used include:

- Recipe metadata: Recipe ID, title, region (continent, country, cuisine), dietary tags (vegan, vegetarian, etc.) .
- Nutritional values: Calories, protein, fat, carbohydrates, etc. (from USDA data).
- Processes: The list of cooking instruction verbs/actions. This is our primary feature. From each recipe’s “Processes” text, we extracted individual protocol terms (verbs) by simple tokenization (splitting on spaces and punctuation) and lemmatization. Non-action words (e.g. prepositions) were discarded. We compiled the set of unique verbs across all recipes.

4.3 Data Cleaning and Preprocessing

Preprocessing steps include:

- Lowercasing: All protocol words were lowercased for consistency.
- Lemmatization: Verbs were lemmatized (e.g. chopping → chop) to merge forms.
- Deduplication: Duplicate terms were removed.
- Filtering: Non-action tokens (measuring words, container names, etc.) were filtered out based on part-of-speech and manual review.

4.4 Feature Engineering

No additional numeric features were engineered beyond the embeddings. However, to enrich our model input, we prepared contextual definitions for each verb. Using lexical resources (Oxford English Dictionary, Cambridge Dictionary, Merriam-Webster) and a culinary resource (Food.com), we collected 2–3 example definitions or descriptions per protocol. These definition texts were optionally concatenated with the verb when generating embeddings (to provide context).

4.5 Dataset Statistics

- Total recipes: 118,000+
- Unique protocols: ~ 270
- Most common protocol terms: mix, stir, cook, chop, bake, etc. (frequency varies by cuisine).
- Regional coverage: recipes from all inhabited continents, with a focus on Asia, Europe, and North America.
- Dietary tags: inclusive of omnivore, vegetarian, vegan, etc., but not directly used in clustering.

SEMANTIC CLUSTERING OF FOOD PREPARATION PROTOCOLS

5.1 Definition of Food Protocols

We define a food preparation protocol as a single cooking action (verb) used in a recipe’s instructions. Examples include chop, grate, boil, whisk, blend, roast, broil, marinate, knead, rest, etc. These verbs encapsulate core operations in the culinary process. As such, protocols are the fundamental “moves” that transform raw ingredients into dishes.

5.2 Protocol Extraction Methodology

From the RecipeDB “Processes” field, we programmatically extracted every verb form. We tokenized each instruction by word boundaries and punctuation, then lemmatized the verbs. For example, the instruction “Peel, chop, and sauté the onions” yields the protocols peel, chop, saute. We aggregated protocols from all recipes into a master list and removed duplicates, resulting in ~270 unique protocol terms. Each protocol was then verified to ensure it represented a cooking action.

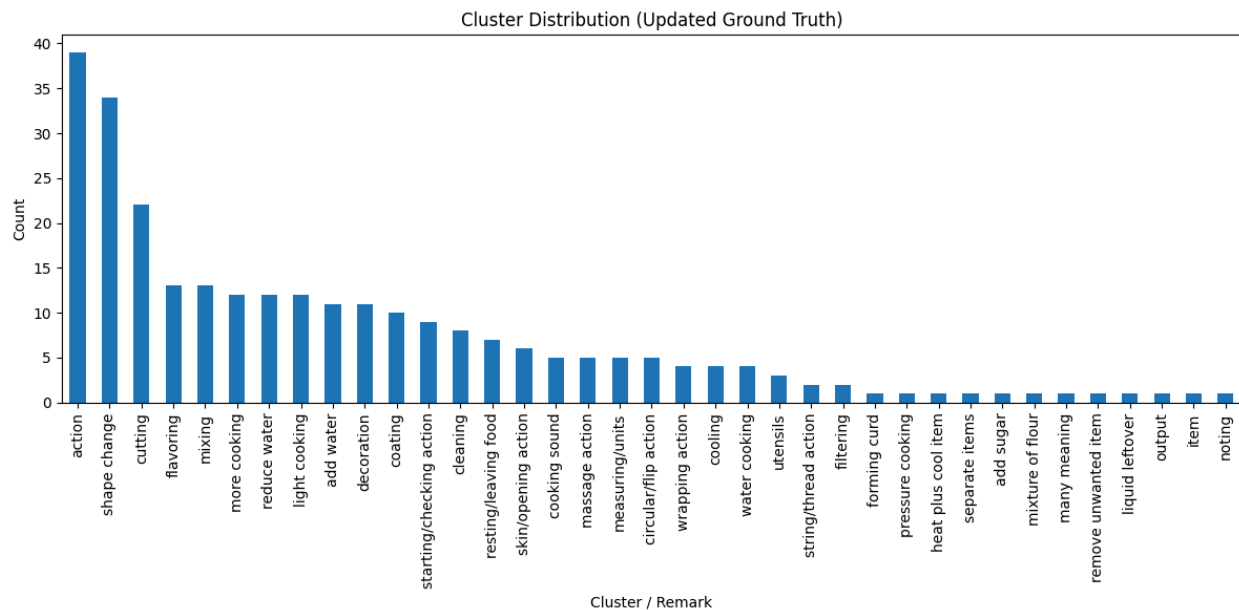
5.3 Protocol Labeling Strategy

To create a ground-truth categorization, each protocol was manually examined. First, we consulted reliable definitions for each verb to understand its meaning in context. For each protocol word, we collected multiple definitions (up to three) from authoritative sources (Oxford English Dictionary, Cambridge Dictionary, Merriam-Webster) and a cooking-specific source (Food.com). This multi-source approach helped clarify nuances (e.g. “baste” meaning vs. “baste” in metalworking) and ensured labeling was grounded in the culinary context.

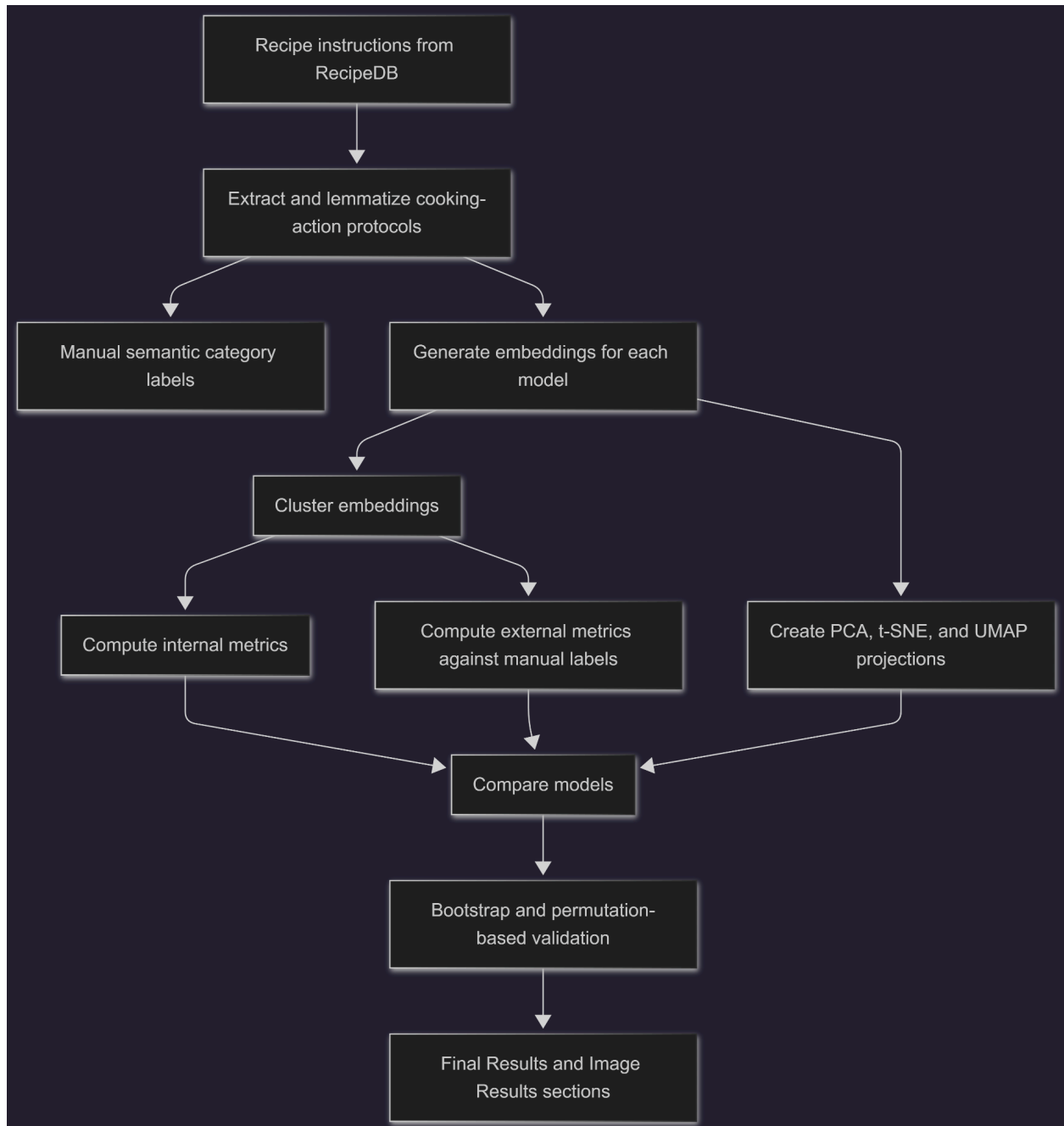
Based on these definitions, we classified each protocol into a semantic subcategory reflecting its function in cooking. For instance, mix, blend, stir, combine, fold were grouped as combination/mixing actions; boil, steam, fry, bake as thermal processing actions; chop, slice, dice as cutting/separation; crush, flatten, melt as transformation (physical change); wash, rinse as cleaning/prep; flip, rotate, twist as directional movements; rest, set, wait as time/holding steps, etc. Words with multiple senses (e.g. jam can mean “to push” or “fruit preserve”) were assigned based on the most relevant cooking sense. In all cases, the chosen label aimed to capture the protocol’s primary role in recipe preparation.

5.4 Ground Truth Construction

From the above labeling, we formed the ground-truth categories. We refined initial broad categories into finer-grained labels where needed. The final taxonomy had around 25-26 protocol categories (e.g. transformation, cutting, mixing, thermal processing, preparation, direction, resting, etc.). Each of the ~270 protocols was assigned exactly one category. This labeled dataset of (protocol, category) pairs serves as the reference for evaluating clustering.



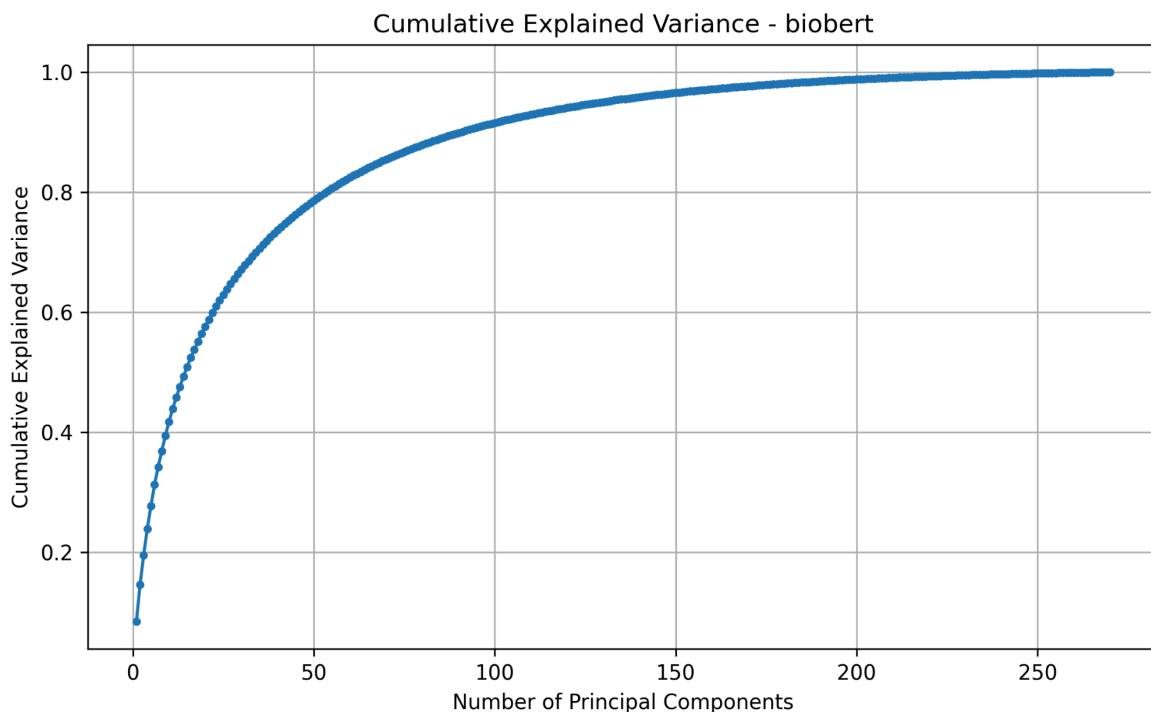
PIPELINE ARCHITECTURE

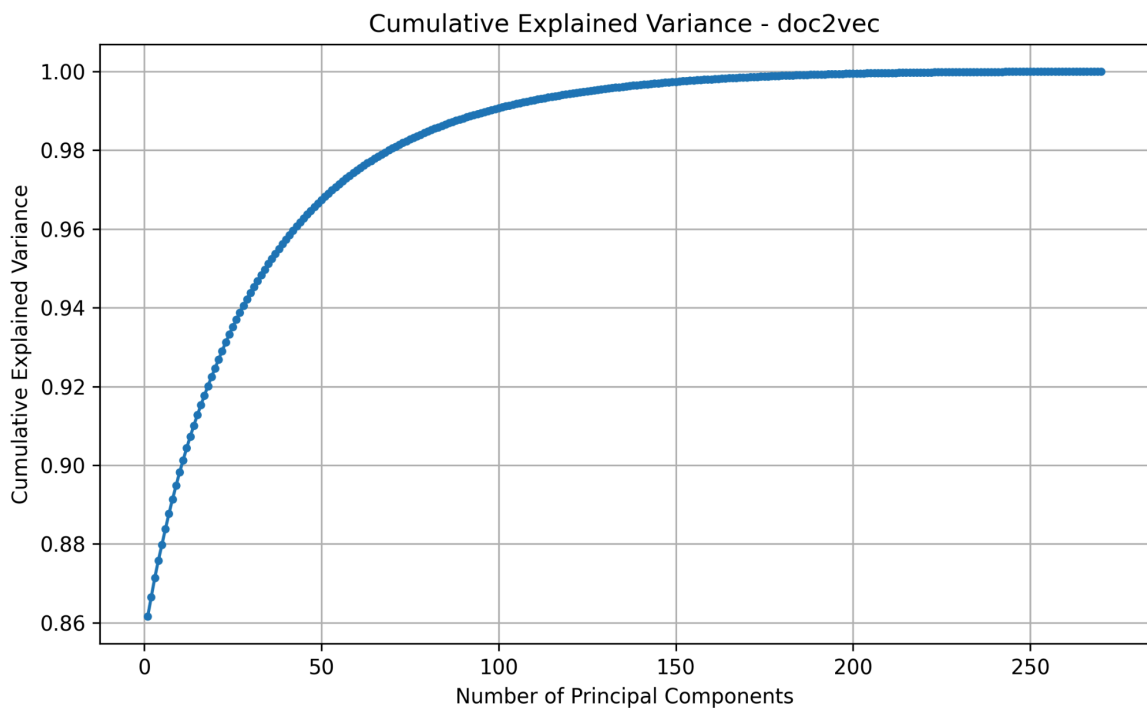
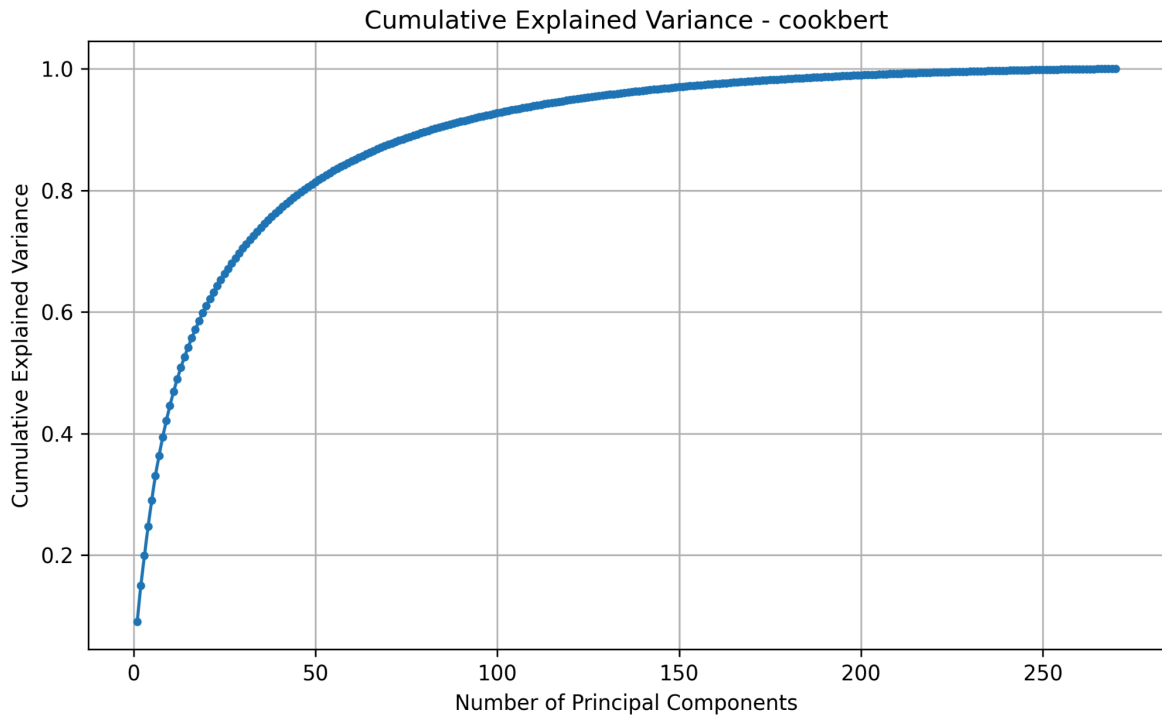


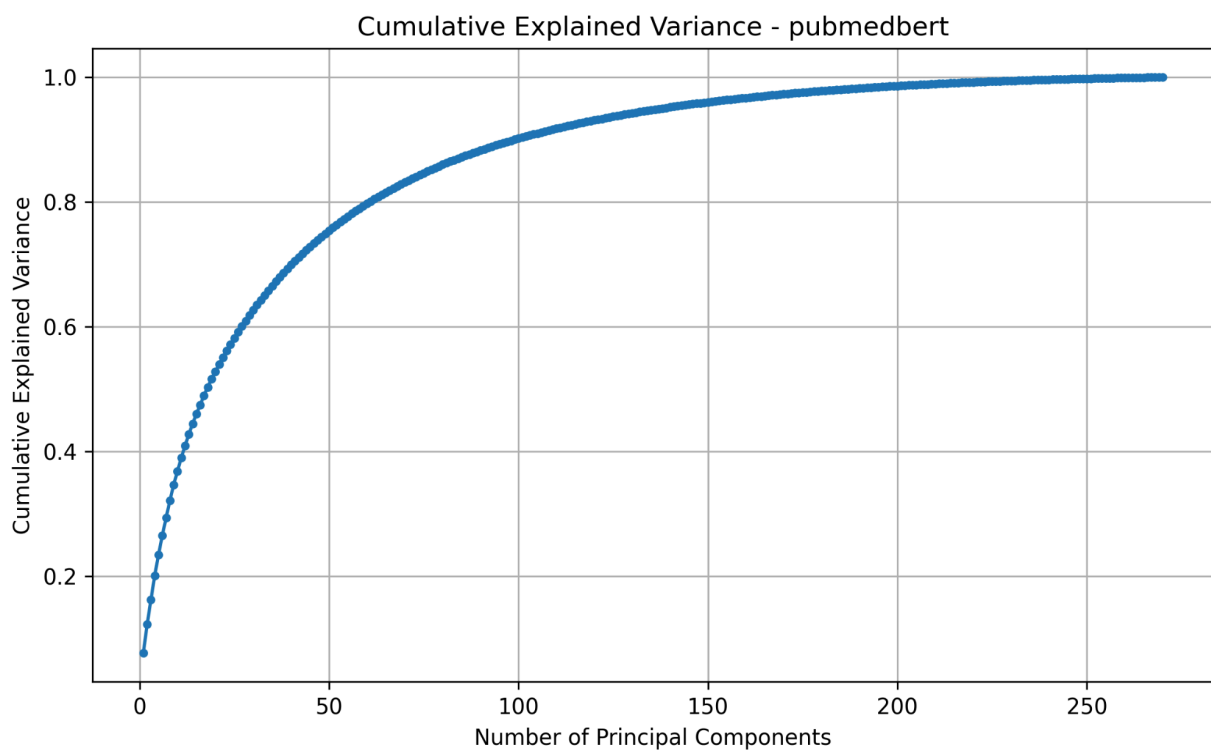
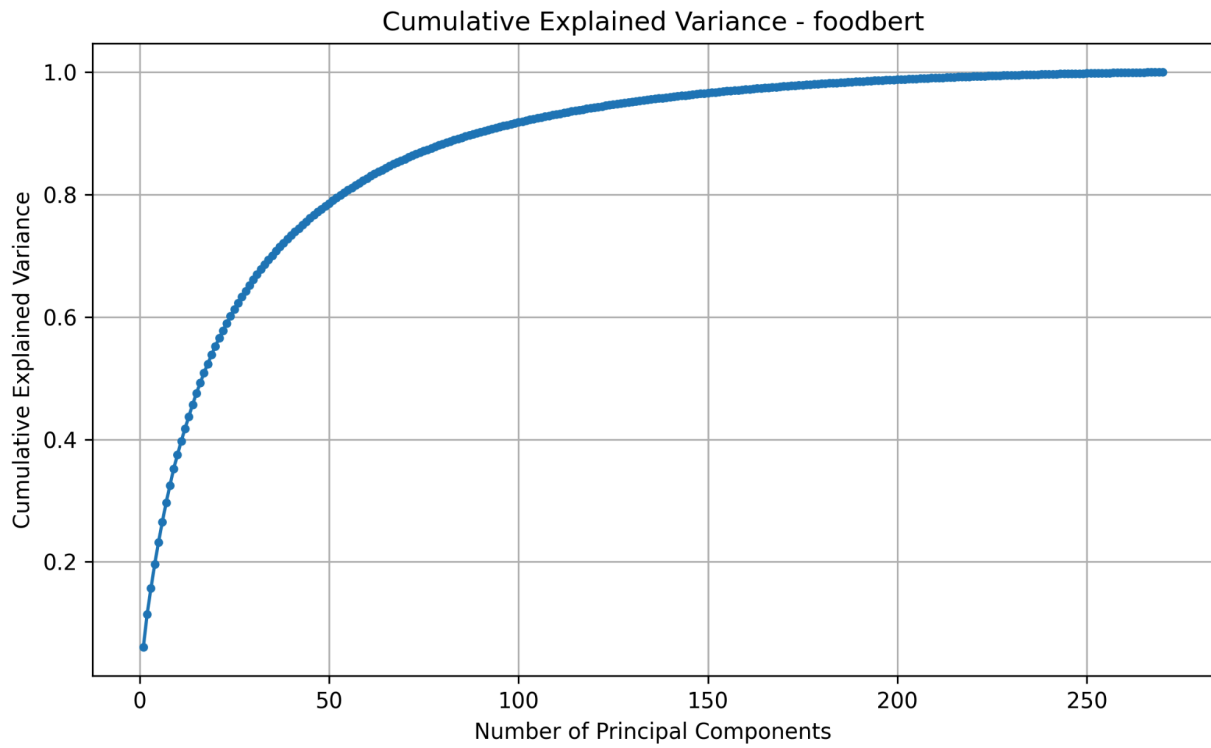
VISUALIZATION AND ANALYSIS

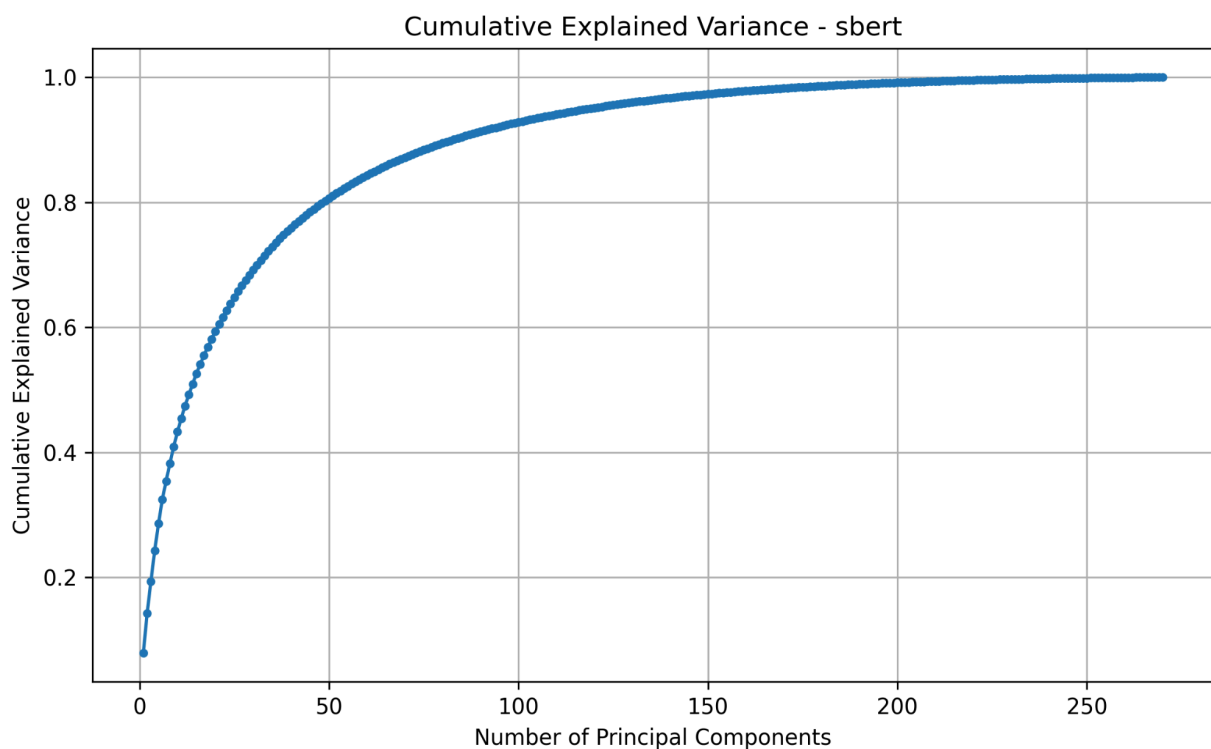
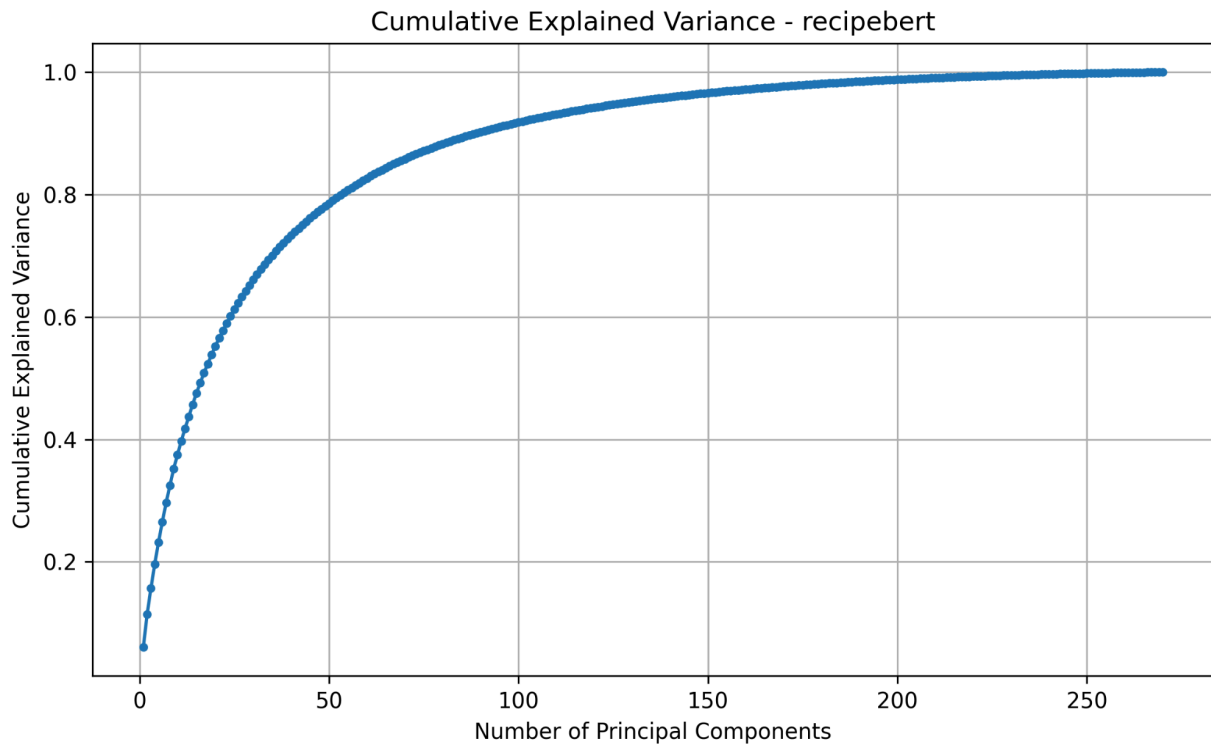
7.1 Cluster Distribution Analysis

The cumulative explained variance plots show how much of the total embedding variance is retained as the number of principal components increases. Across the transformer-based models, the variance increased gradually, indicating that the semantic information was distributed across many dimensions rather than concentrated in only a few. This suggests that these embedding spaces preserve richer and more complex semantic structure. The SBERT and CookBERT plots showed smoother accumulation of variance, which is consistent with their stronger semantic clustering performance.







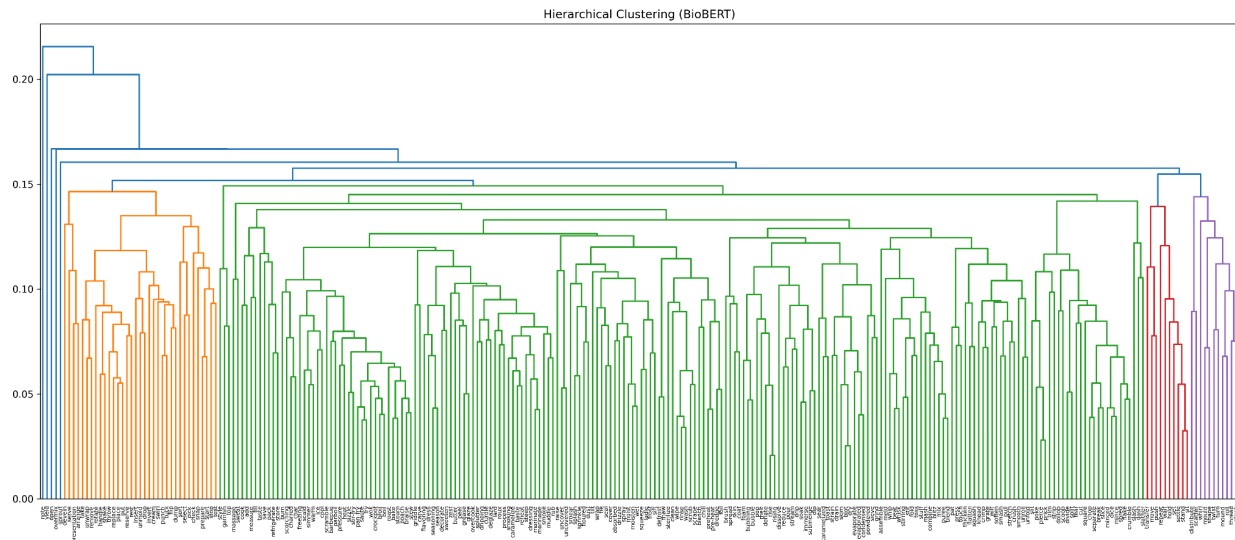


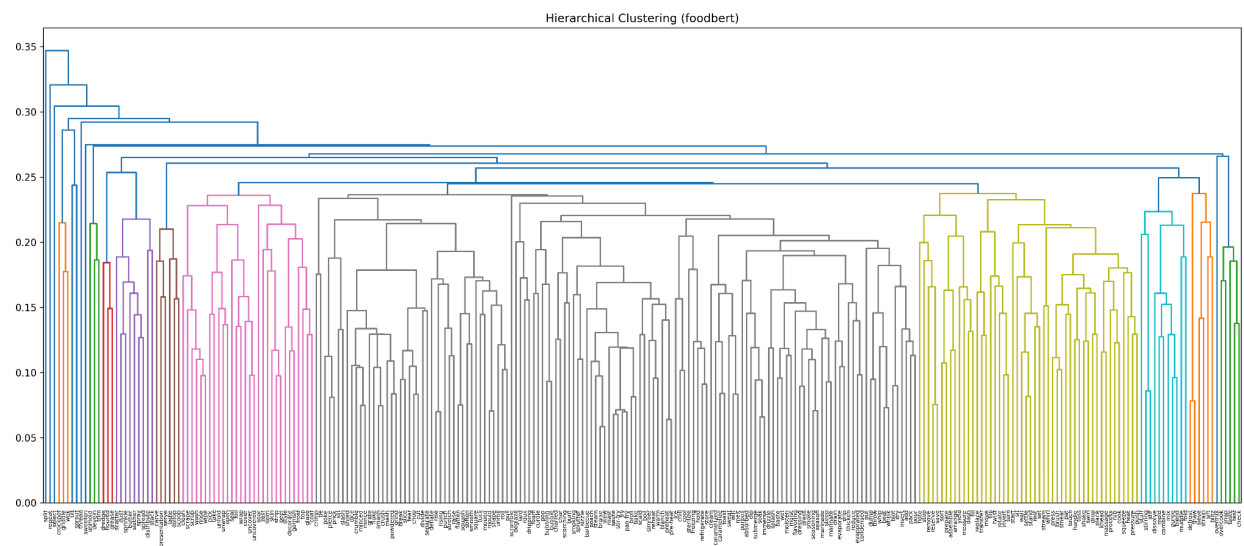
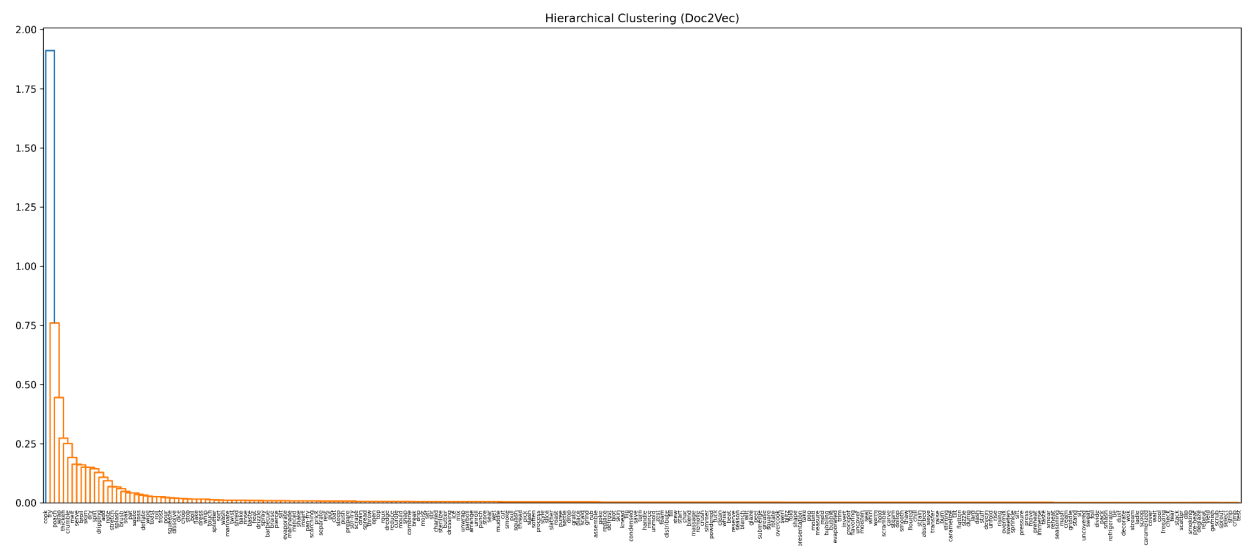
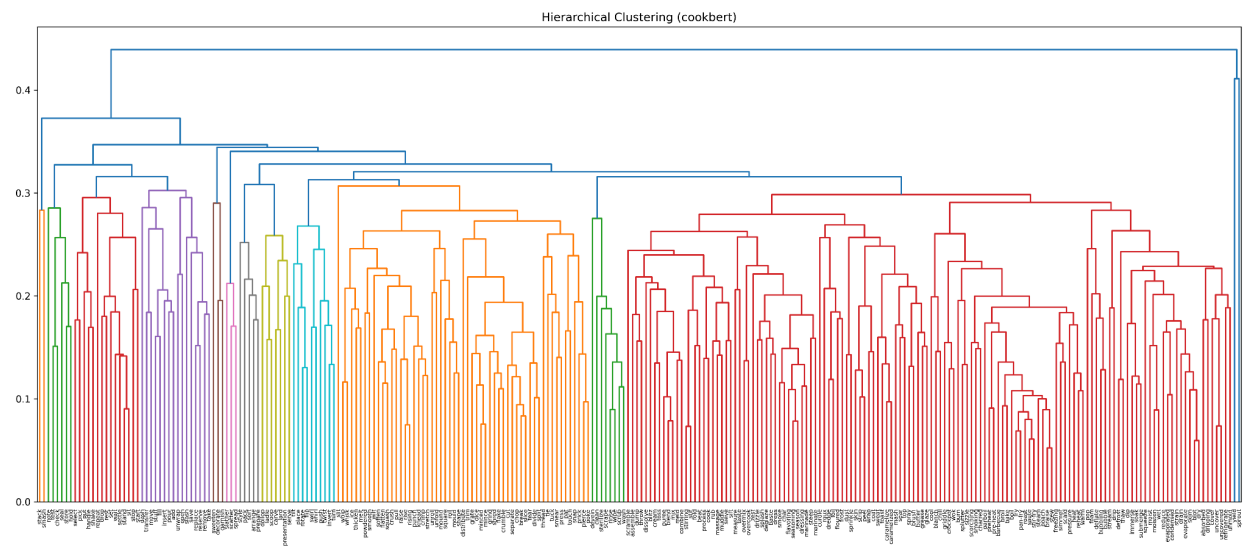
In contrast, **Doc2Vec** reached near-total explained variance much faster than the other models, meaning that its representation was

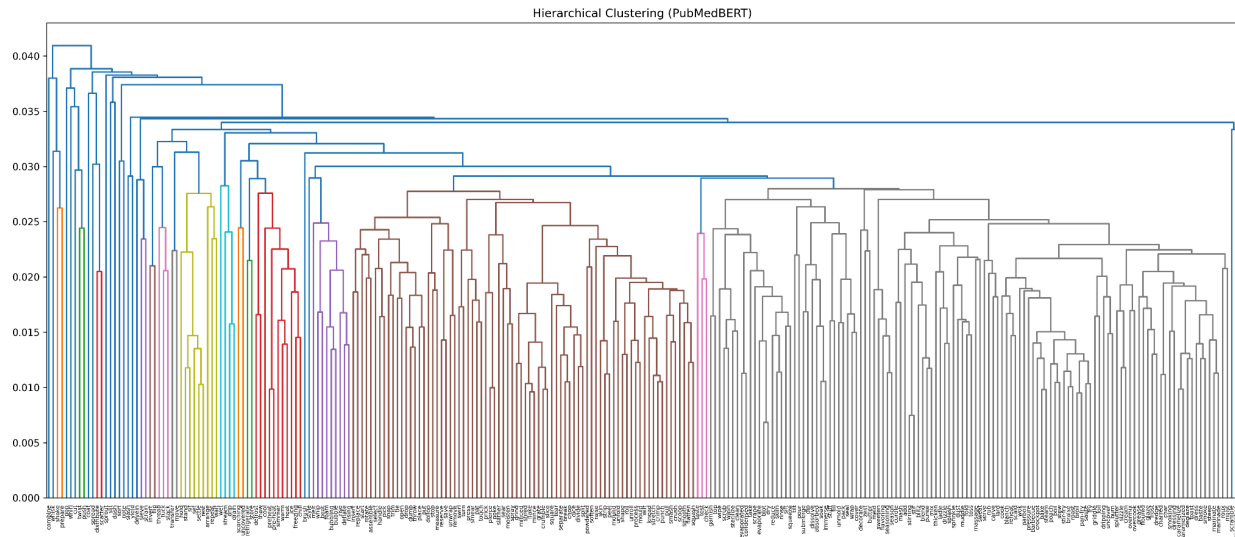
concentrated in fewer dominant components. While this may indicate simpler geometric structure, it also suggests reduced semantic richness. This explains why Doc2Vec could produce compact clusters while still failing to align with the actual category labels. Thus, the cumulative variance analysis supports the conclusion that **SBERT and CookBERT preserve more useful semantic information**, whereas Doc2Vec forms a lower-dimensional but less semantically meaningful embedding space.

7.2 Dendrogram Analysis

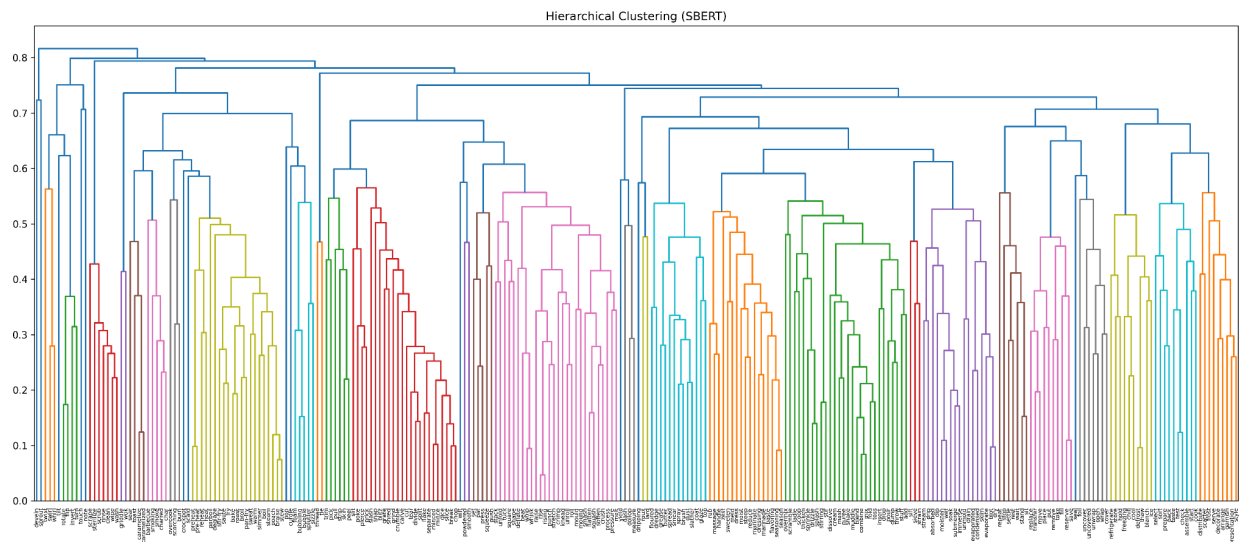
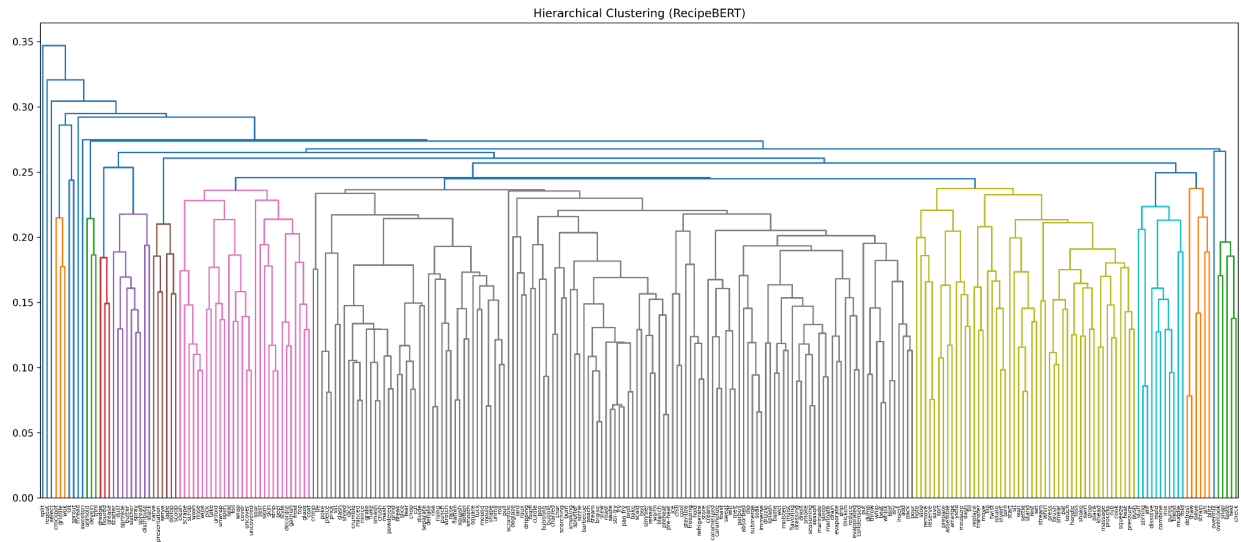
The dendrograms provide a hierarchical view of how protocol terms are merged during agglomerative clustering. A good dendrogram should show meaningful branches where semantically similar actions are grouped together and unrelated actions merge only at larger distances.





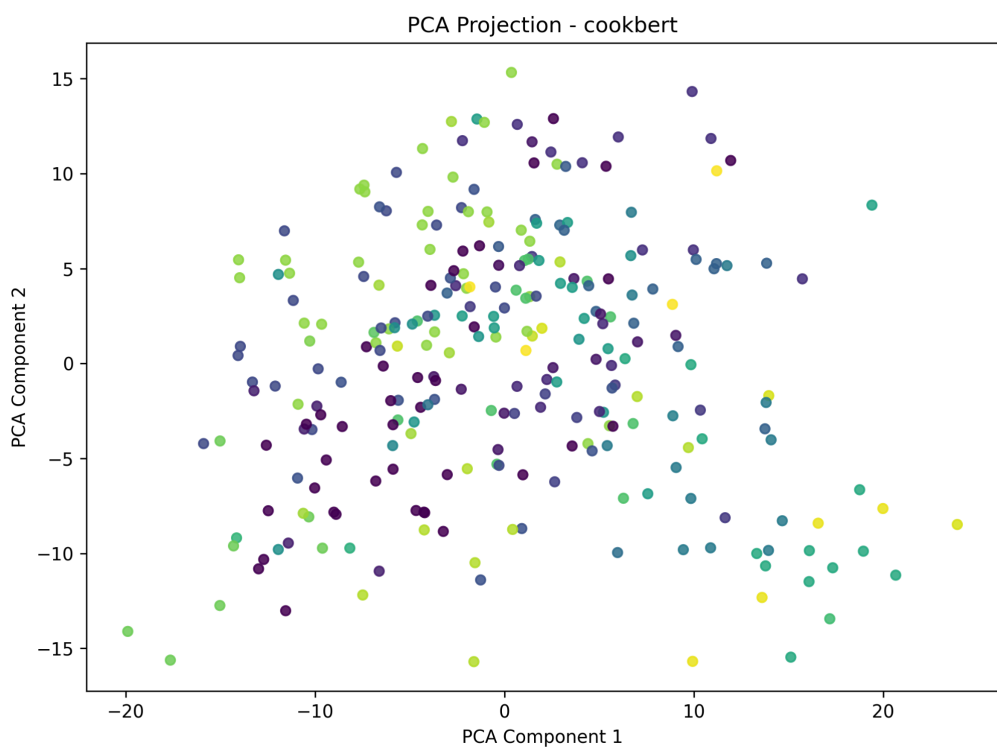
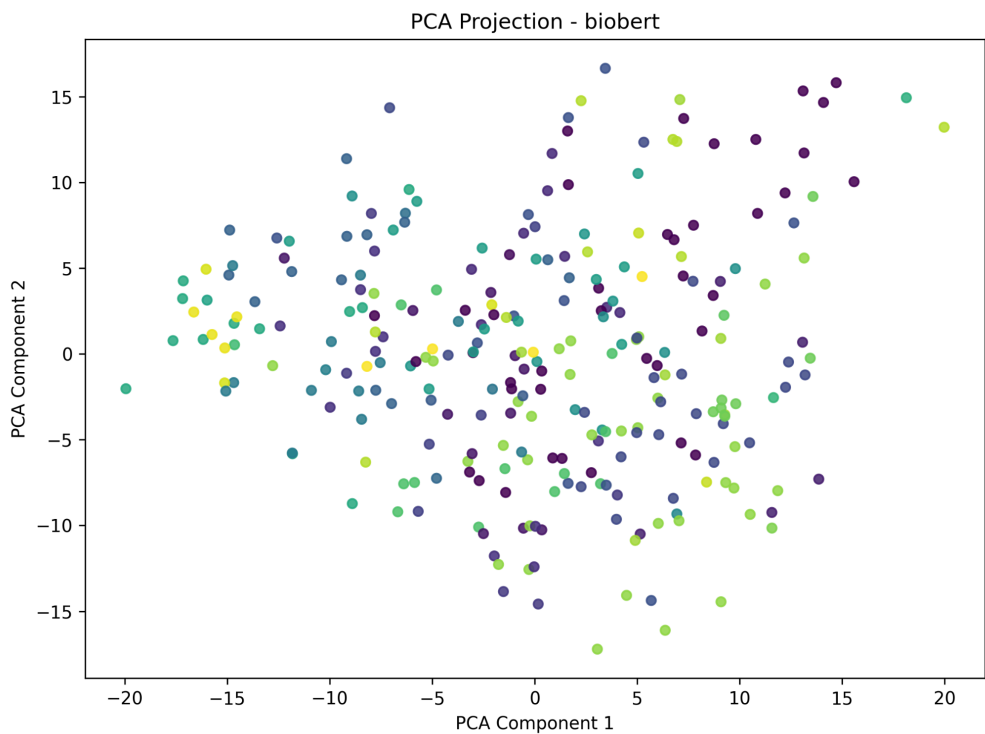


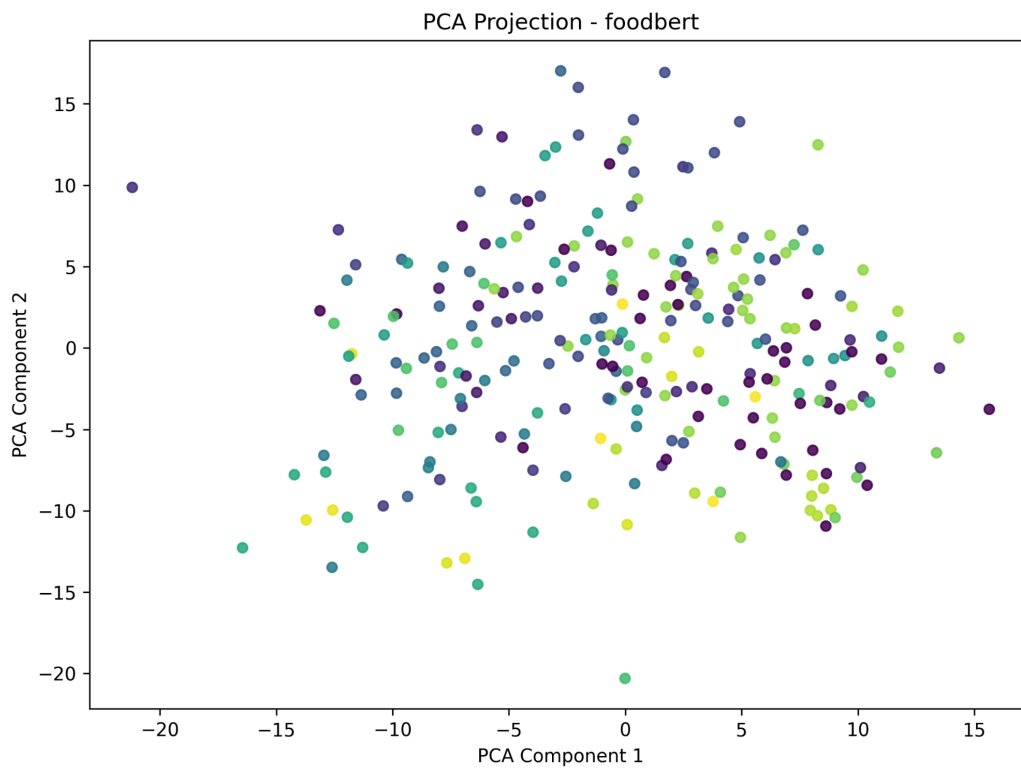
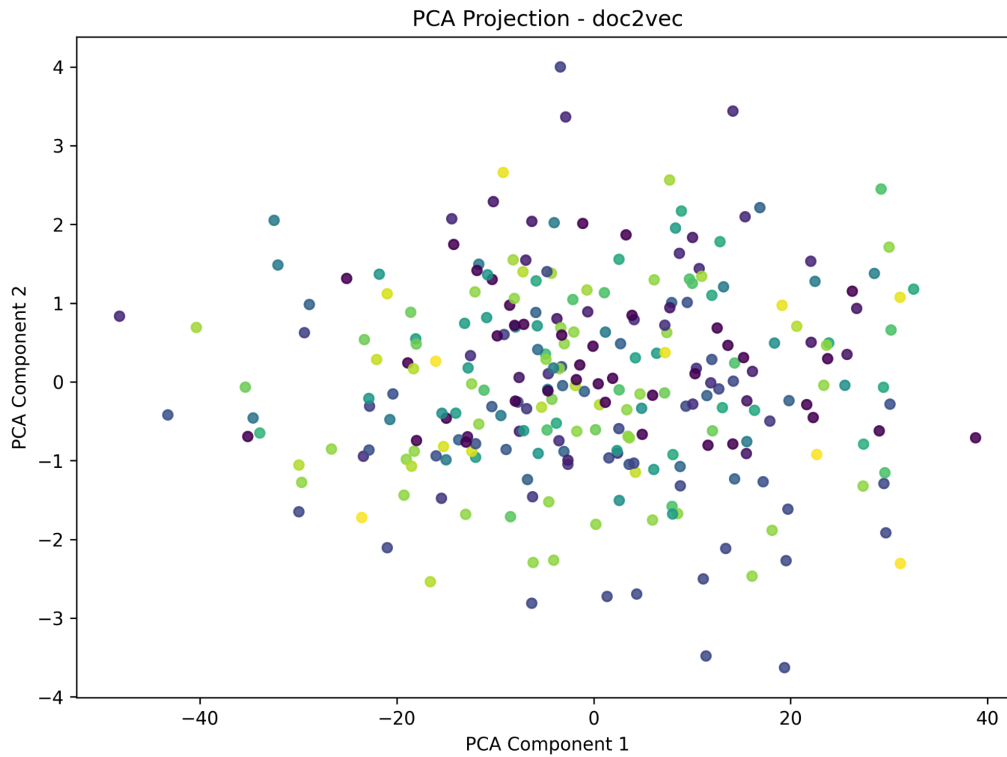
Among the evaluated models, **SBERT** and **CookBERT** produced more interpretable hierarchical patterns, with clusters forming in a more balanced and semantically plausible manner. Their branch structures suggest that related cooking actions remained grouped for longer stages of the clustering process, which is consistent with their relatively better ARI scores. These models therefore appear to preserve semantic similarity more effectively in hierarchical clustering. By contrast, **Doc2Vec** showed an overly compressed dendrogram structure, where many items appeared to merge rapidly without preserving fine-grained semantic distinctions. This indicates that although the model separated the data strongly at a coarse level, it failed to capture the nuanced relationships needed for meaningful semantic taxonomy construction. The dendrogram analysis therefore supports the quantitative results by reinforcing that **SBERT provides the most semantically meaningful hierarchical organization of food preparation protocols.**



7.3 PCA Visualization

The PCA visualizations present a linear two-dimensional projection of the protocol embeddings and help in observing the global structure of the semantic space. The PCA plots show that most models exhibit some overlap between categories, which indicates that the clustering task is inherently complex and not perfectly separable in a simple linear projection.

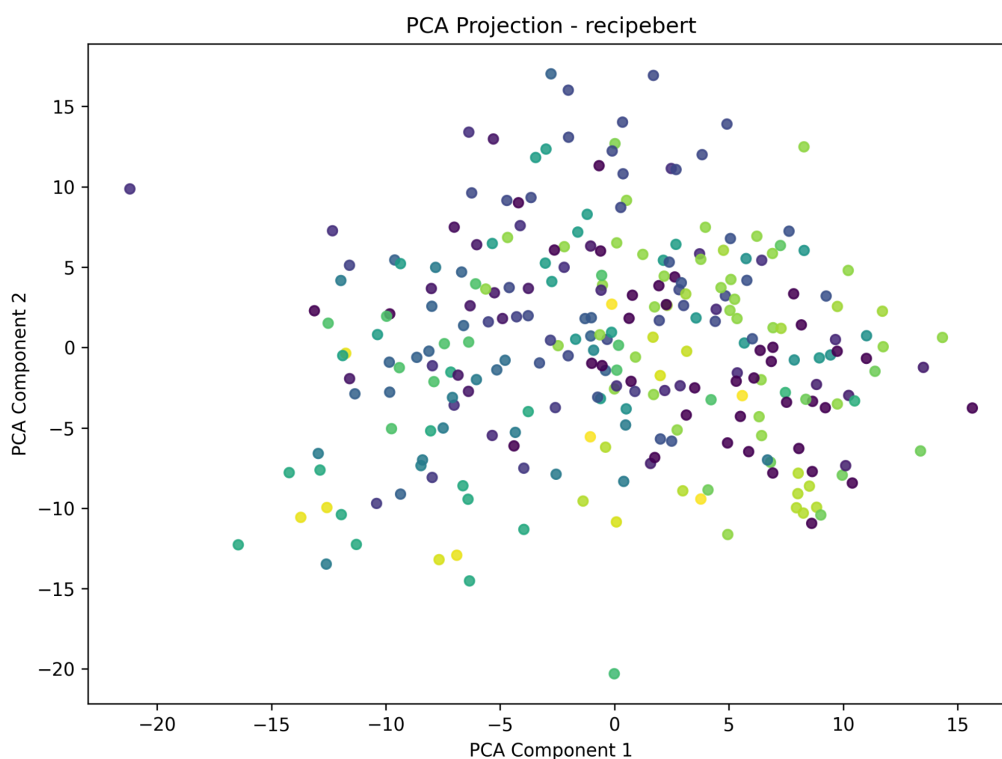


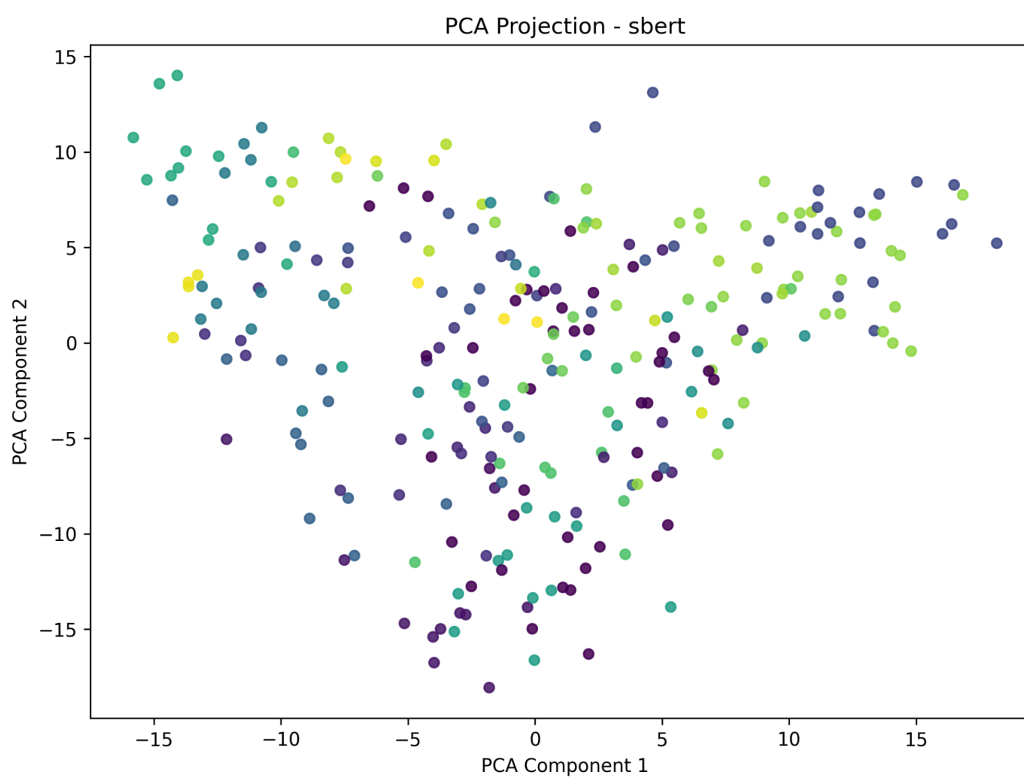
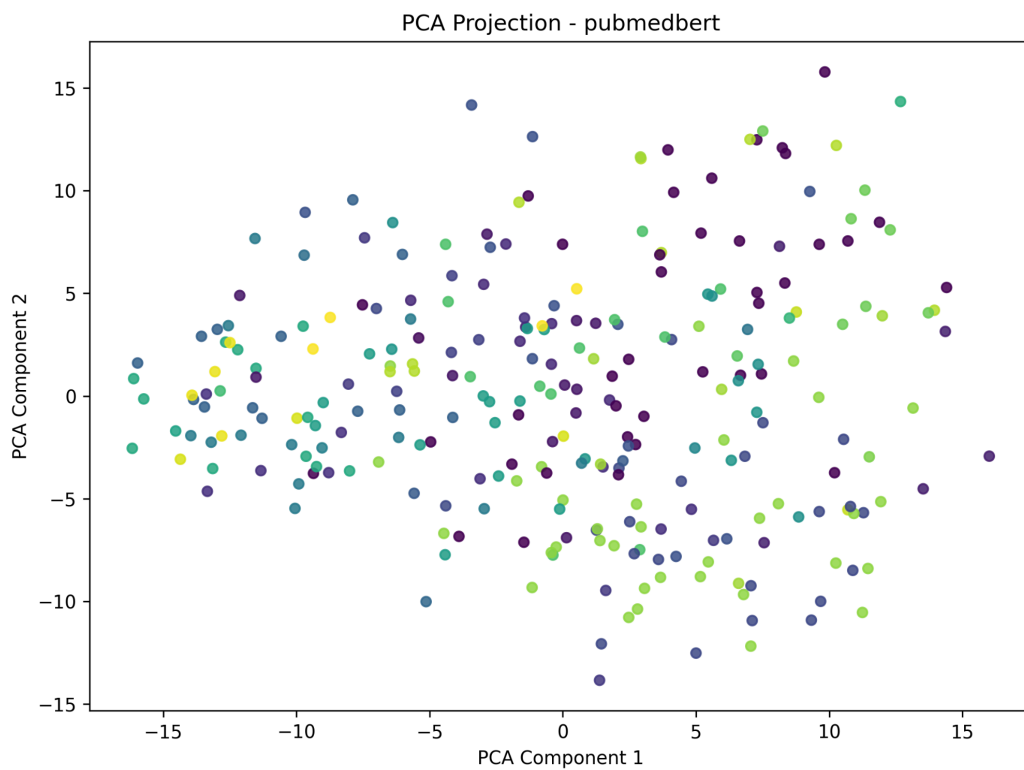


Among all models, **SBERT** displayed relatively better global organization, with a more structured spread of points and visible

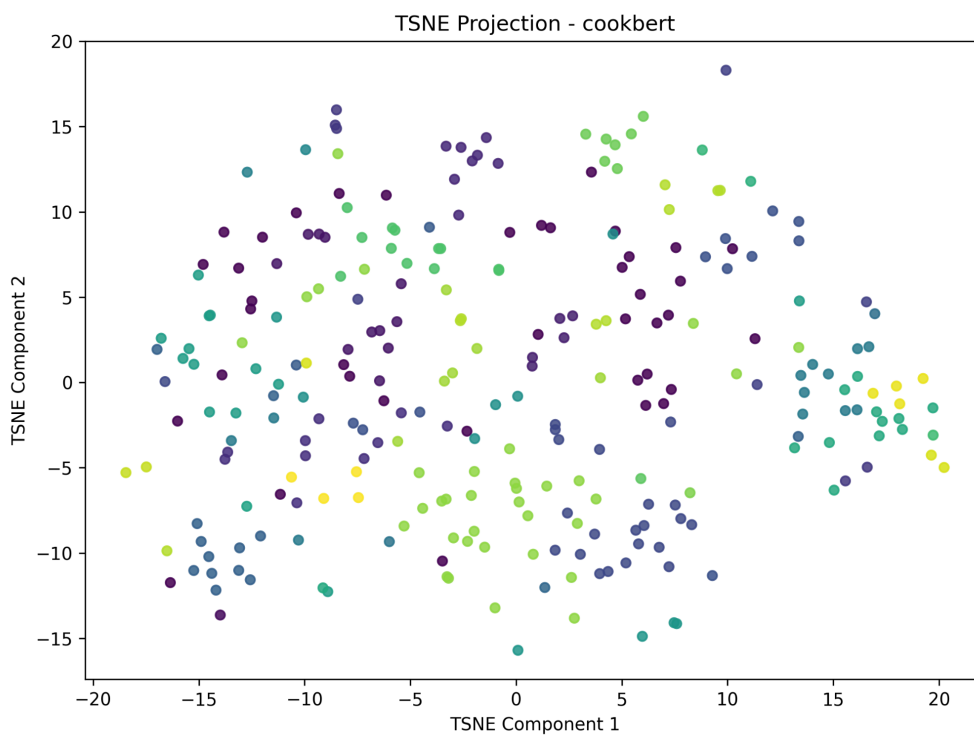
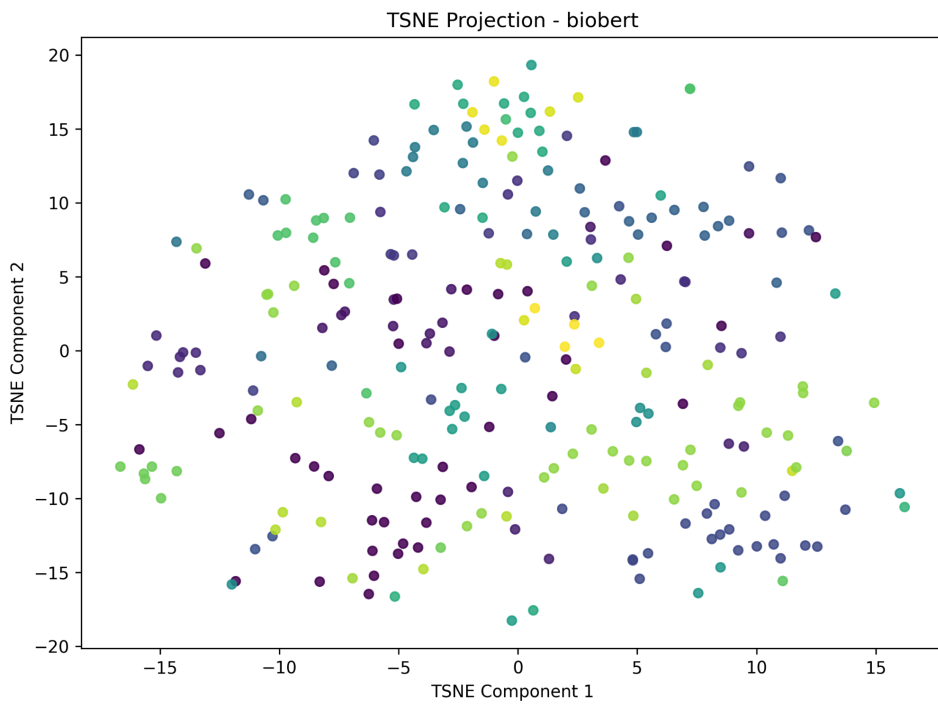
grouping tendencies compared to the weaker baselines. **CookBERT** also showed some degree of separation, though the clusters were less distinct and more intermixed than in SBERT. These observations align with their stronger ARI scores and suggest that both models encode meaningful semantic variation in the embedding space.

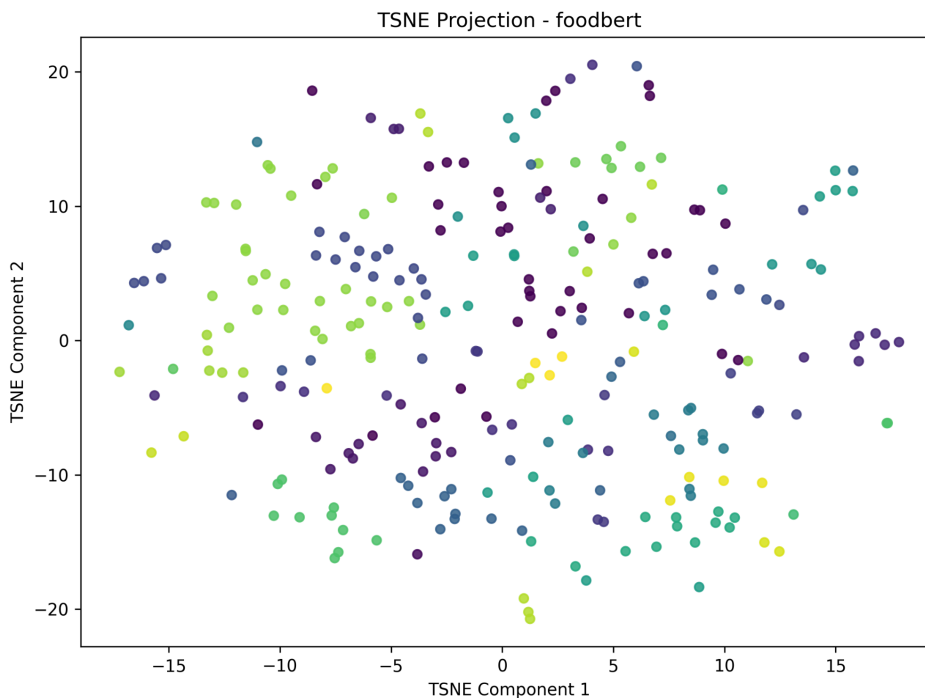
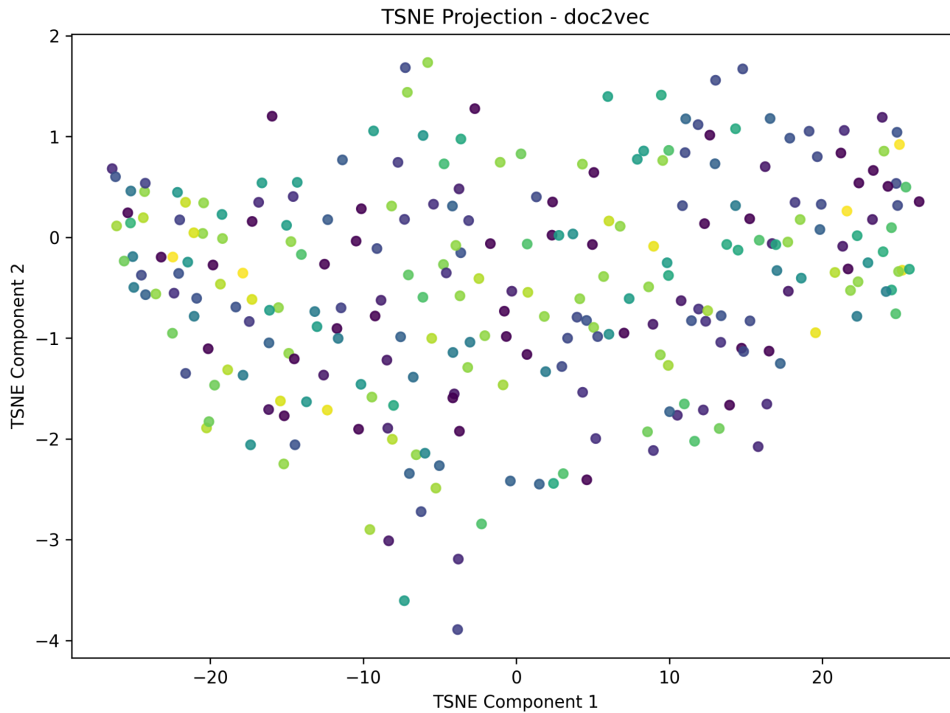
On the other hand, **RecipeBERT**, **FoodBERT**, **BioBERT**, and **PubMedBERT** showed more scattered and overlapping point distributions, making it difficult to identify clear semantic groups. **Doc2Vec** appeared geometrically compact, but its PCA projection did not reveal category-aligned separation. Therefore, the PCA analysis indicates that **SBERT provides the most coherent global representation**, while the weaker models fail to organize the protocols into clearly distinguishable semantic regions.





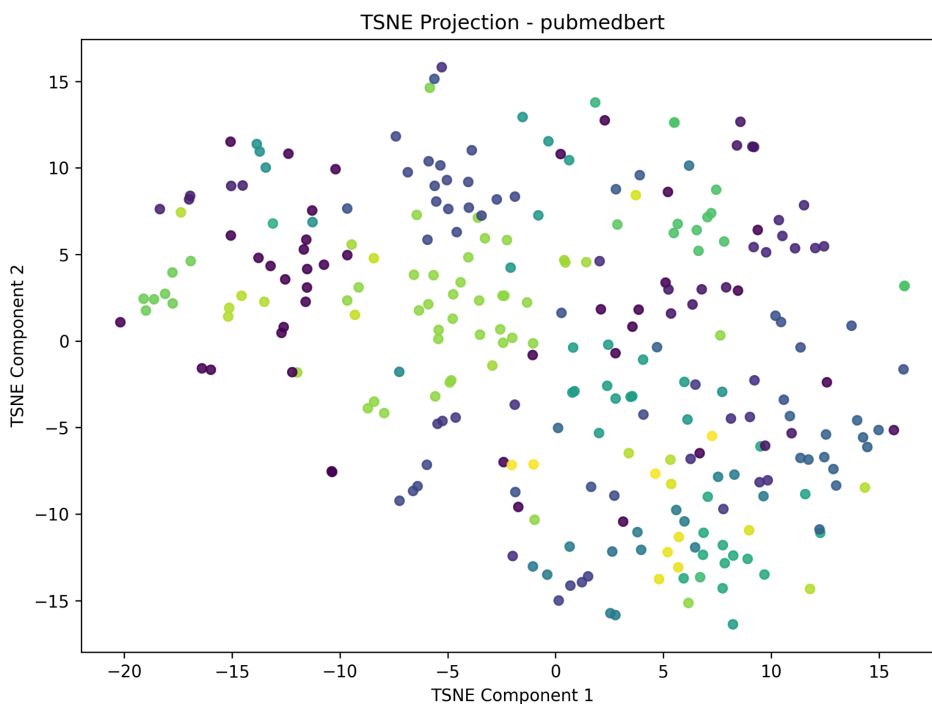
7.4 t-SNE Visualization





The t-SNE plots provide a nonlinear visualization of local neighborhood structure in the embedding space. Unlike PCA, t-SNE emphasizes local similarity, making it useful for identifying whether semantically related protocols tend to form tight local neighborhoods.

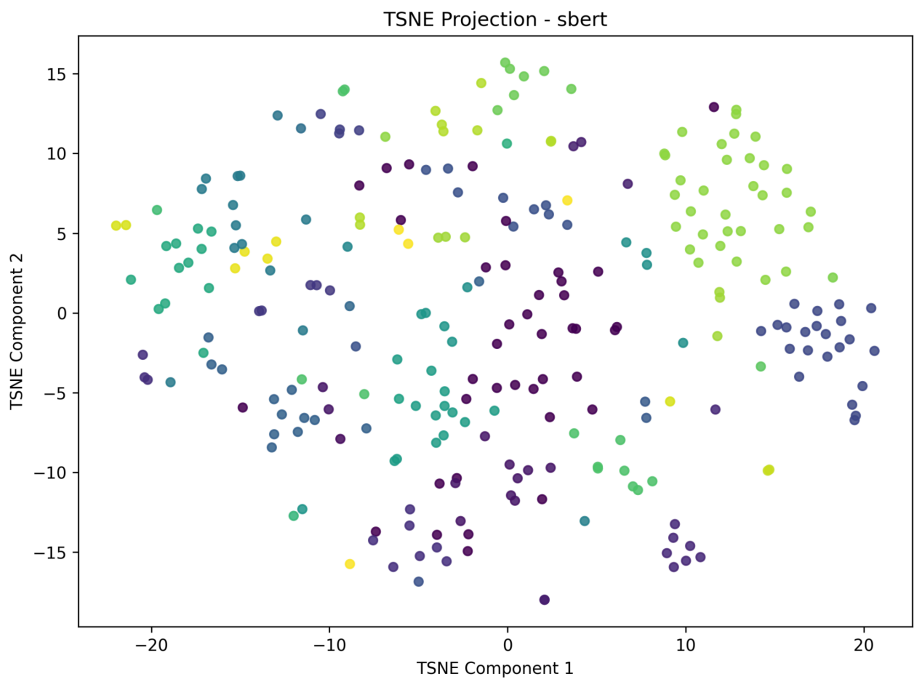
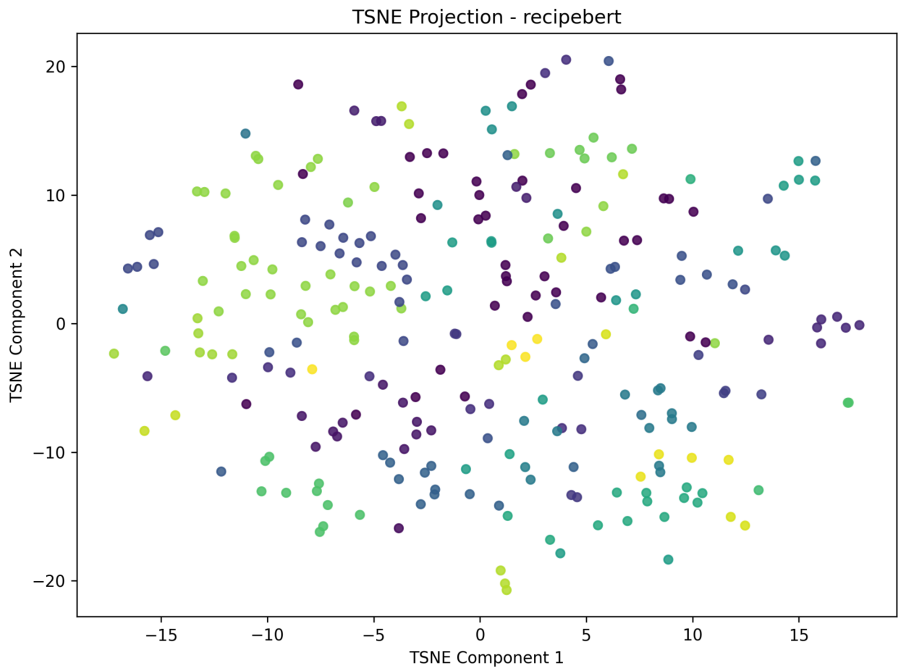
The t-SNE projections reveal that the transformer-based models differ substantially in how well they preserve these local relationships.



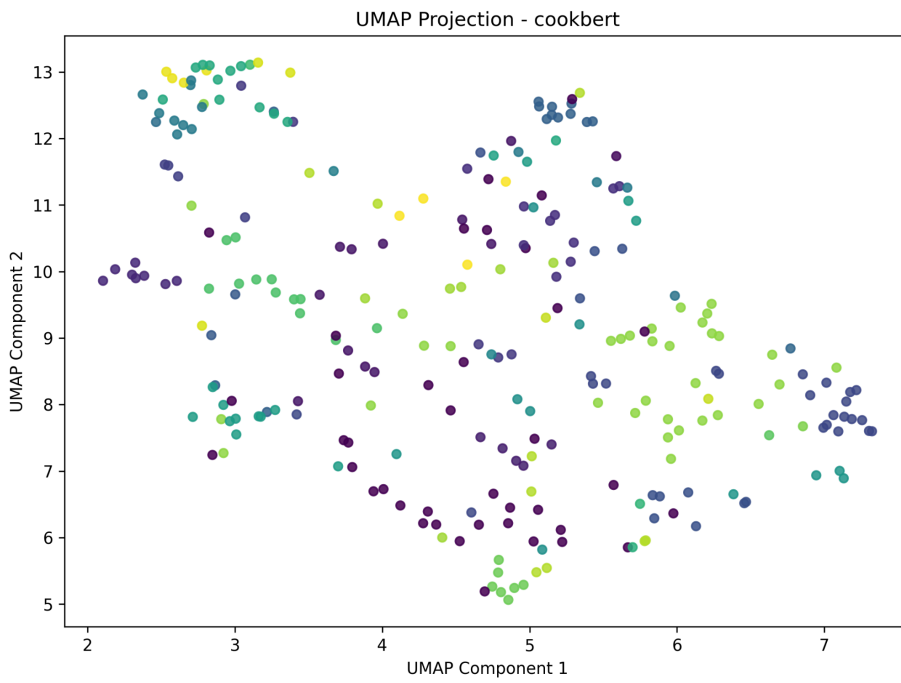
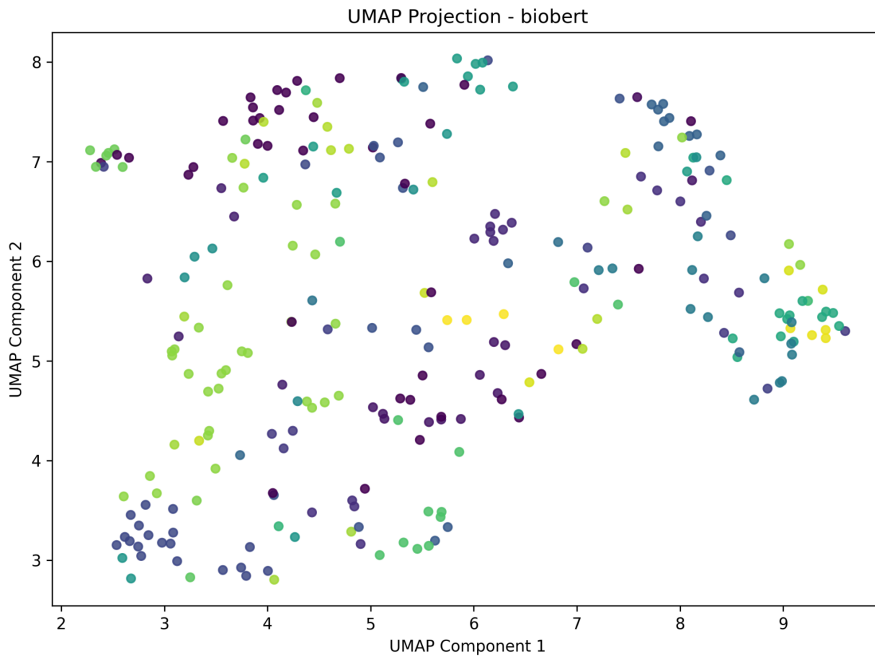
SBERT showed comparatively better local grouping, with several regions of denser point organization than the other models.

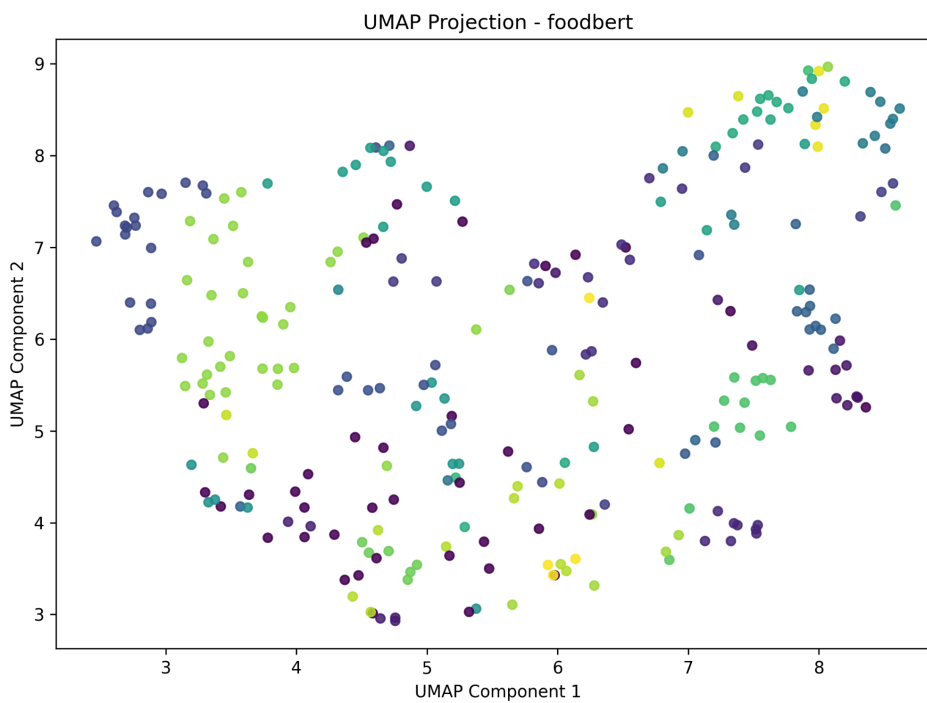
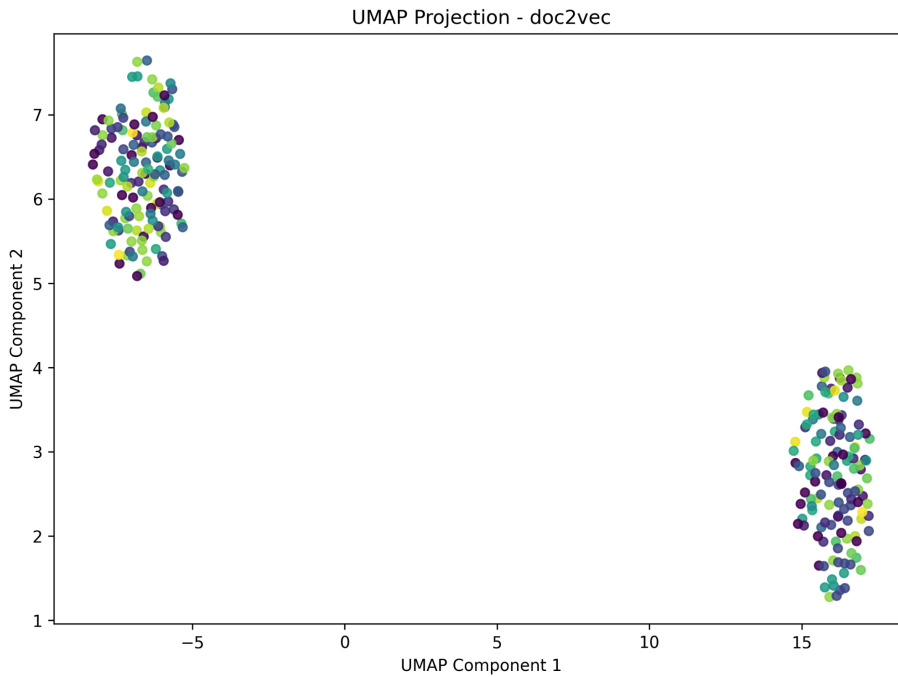
CookBERT also demonstrated moderate neighborhood preservation, though its clusters were more diffuse and less well-separated than those of SBERT. These observations suggest that these two models capture local semantic similarities more effectively than the remaining baselines.

The other models, especially **PubMedBERT** and **FoodBERT**, showed heavy overlap and weak local structure, indicating poorer semantic grouping. **Doc2Vec** again displayed visually noticeable separation, but this separation was not semantically meaningful when compared against the ground-truth categories. Therefore, the t-SNE analysis supports the overall conclusion that **SBERT yields the most semantically reliable local structure among all tested embeddings.**



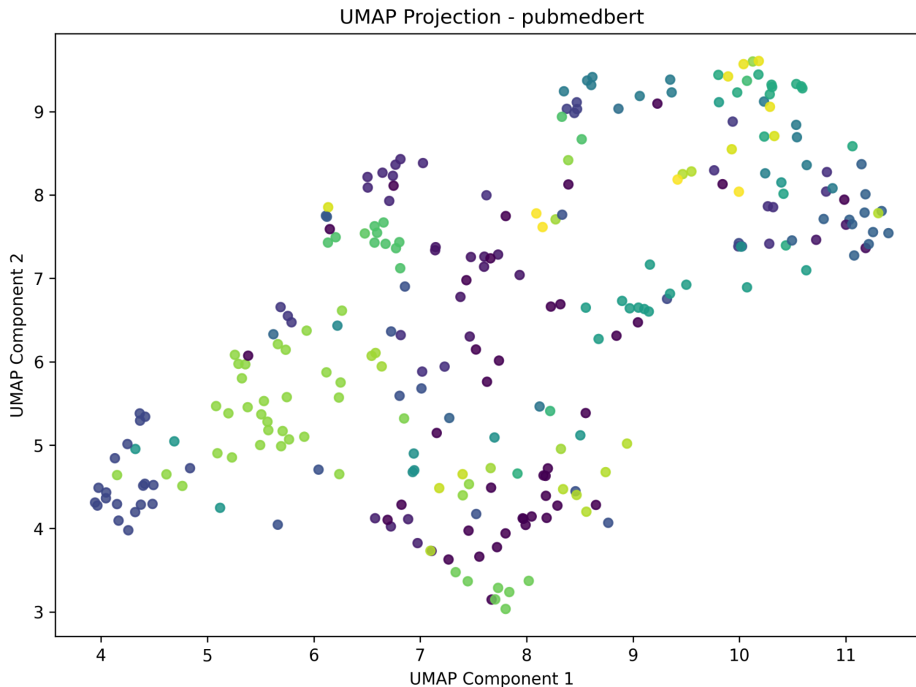
7.5 UMAP Visualization





The UMAP projections provide another nonlinear view of the embedding space and are especially useful for examining both local grouping and broader manifold structure. The UMAP plots on pages 25–29 were among the clearest visualizations in the study and offered

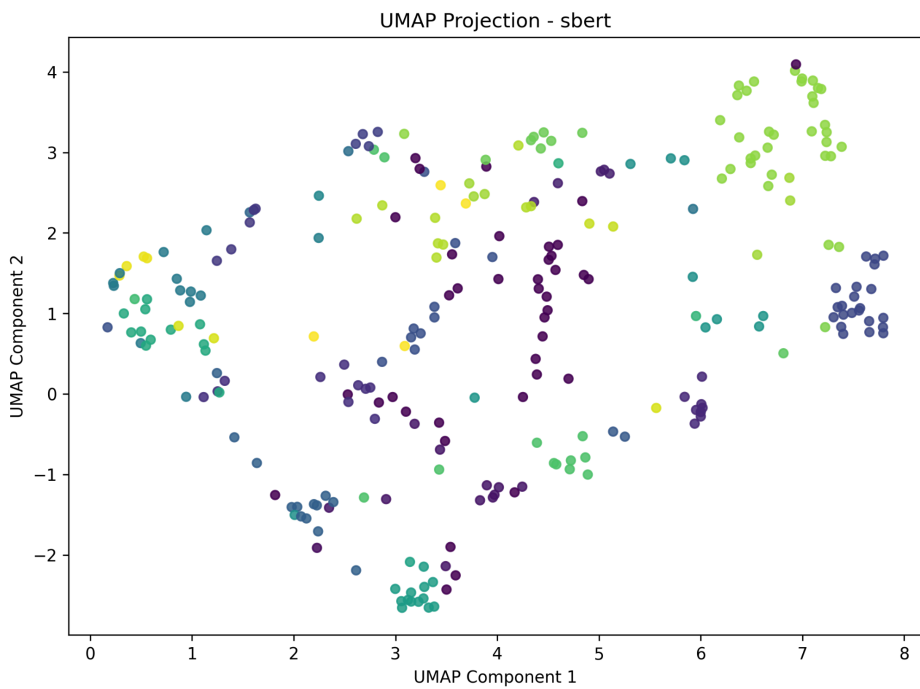
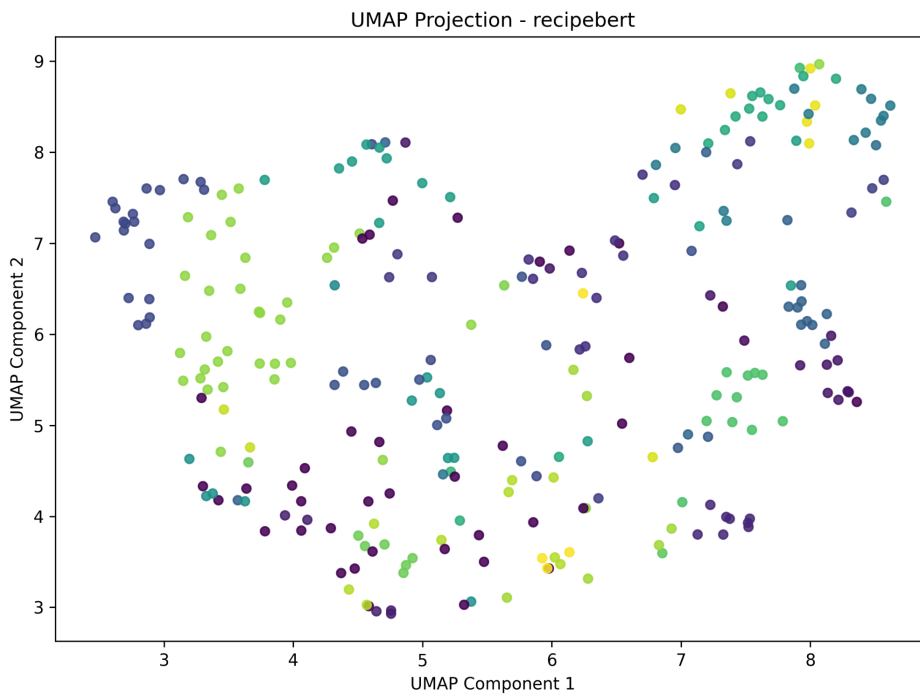
strong qualitative support for the quantitative evaluation results.



Among all models, **SBERT** showed the most meaningful organization in the UMAP space, with several visibly distinct regions that suggest better separation of semantic protocol types. **CookBERT** also formed partially separated groups, though the clusters remained less well-defined than those seen in SBERT. This again reflects the ranking observed in ARI-based evaluation, where SBERT clearly outperformed the remaining models.

A striking observation was the UMAP projection of **Doc2Vec**, which showed very strong visual separation into major groups. However, this apparent success was misleading because the ARI score showed almost no agreement with the human-labeled categories. Thus, the UMAP results highlight an important lesson of this study: **visual separation alone does not guarantee semantic correctness**. Even though Doc2Vec produced highly distinct clusters in projection space, those clusters did not represent the intended food preparation protocol taxonomy. Overall, the UMAP analysis confirms that **SBERT offers**

the best balance of visual separability and semantic alignment,
making it the most suitable model for this task.



CLUSTERING RESULT

Model	Silhouette Score	ARI Score	ARI Interpretation
SBERT	0.1655	0.5360	Best alignment
RecipeBERT	0.0586	0.1550	Low
CookBERT	0.1228	0.2810	Moderate
BioBERT	0.0932	0.1678	Low
FoodBERT	0.0587	0.1550	Low
PubMedBERT	0.0212	0.0686	Very Low
Doc2Vec	0.6333	-0.0036	Poor (random alignment)

The Adjusted Rand Index (ARI) was used as the primary evaluation metric to measure how well the generated clusters aligned with the manually constructed ground-truth semantic categories of food preparation protocols. Among all evaluated embedding models, SBERT achieved the highest ARI score of 0.5360, indicating the strongest agreement with the human-labeled taxonomy. This shows that SBERT was the most effective model for capturing the semantic relationships between culinary action verbs and grouping them into meaningful categories. The relatively higher ARI suggests that SBERT embeddings preserved semantic similarity in a way that was consistent with expert-defined protocol classes.

CookBERT obtained the second-best ARI score of 0.2810, showing moderate alignment with the ground-truth labels. Although its performance was notably lower than SBERT, it still demonstrated some ability to capture domain-relevant semantic structure. This suggests that a cooking-domain-oriented model can offer useful

clustering behavior, but in this case it was not as robust as SBERT for semantic grouping.

BioBERT, RecipeBERT, and FoodBERT produced comparatively lower ARI scores, indicating weak agreement with the manual labels. Their results suggest that these models were able to capture limited semantic structure, but the resulting clusters were not sufficiently aligned with the intended culinary protocol categories. PubMedBERT performed even worse, with a very low ARI, which implies that a biomedical-domain model does not transfer effectively to the food protocol clustering task.

A particularly important observation is the performance of Doc2Vec, which produced an ARI score of -0.0036. This indicates almost no meaningful agreement with the ground-truth categories despite showing strong internal compactness in other metrics. Therefore, although Doc2Vec formed visually separated groups, these clusters did not correspond to the actual semantic categories defined in the dataset. Overall, the ARI-based evaluation confirms that SBERT is the best-performing embedding model for semantic clustering of food preparation protocols, with CookBERT as a distant second.